# Parth Mhaske

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## Education

**University of Maryland** — College Park, MD
***B.S. in Computer Science and Applied Mathematics (GPA: 3.7)*** — ***Spring 2028***

- **Relevant Coursework:** Algorithms, Data Structures, Advanced Linear Algebra, Probability Theory, Computer Systems, Functional Programming Languages
- **Clubs & Organizations:** Smith Investment Fund, Competitive Programming Club, Club Cycling Team
- **Awards:** 2x AIME; Science Olympiad Nationals 3rd; FRC World Championship Autonomous Award
- **Publications:** Mhaske, P., and Claudel, C., "Suspension Parameter Optimization in a Paramotor UAV Using Monte Carlo Analysis," AIAA 2026-4336, AIAA AVIATION 2026 Forum, San Diego, CA, June 2026. https://doi.org/10.2514/6.2026-4336

## Experience

**Corsha** — May 2026 - Present
*Software Engineering Intern* — *Vienna, VA*

- Built Go service that auto-detects client OS/arch at handshake, removing per-client manual configuration.
- Architected a log-ingestion service that buffers spike traffic to prevent core-API degradation and data loss.

**Architect Labs (Prev. Nuntius YCombinator 'S25)** — Nov 2025 - May 2026
*Software Engineer* — *(Remote)*

- Designed and implemented novel agentic environments with synthetic data generation and automatic evaluation pipelines.
- Built custom reward system for LLM tool-call correctness; surfaced 88% frontier-model failure on 100+ multi-turn benchmarks.
- Conducted penetration testing and failure analysis on AI models, uncovering edge cases that informed hardening strategies.

**Smith Investment Fund** — Oct 2025 - Present
*Quantitative Analyst* — *College Park, MD*

- Built a C++ Crypto.com HFT (Boost.Beast, lock-free MPSC, cache-aligned trades) with latency of ∼110 ns p99 tick-to-order.
- Designed an event-driven StrategyBase that each strategy subclasses to emit signals, validated on a JSONL replay backtester.

**Department of Astronomy, University of Maryland** — Jan 2026 - Present
*Undergraduate Researcher* — *College Park, MD*

- Reduced MSTM dust-scattering compute time from hours to seconds ($> 1000\times$ speedup) using a Gaussian Process surrogate.
- Mapped polarization profile of interstellar comet 3I/ATLAS across 4 wavelength bands along tail and from core.

**Computational Social Dynamics Lab, University of Maryland** — Jan 2026 - Present
*Undergraduate Researcher* — *College Park, MD*

- Extended Ciampaglia et al. (2018) to study popularity-drive bias across 5 cultural-market datasets and 30M ratings.
- Evaluated popularity-bias models under personalization, manipulation, and dynamic-quality regimes.
- Used Approximate Bayesian Computation to calibrate popularity-bias simulations to empirical ranking data.

**MASS Lab, University of Texas at Austin** — Apr 2024 - Jan 2025
*Research Assistant* — *Austin, TX*

- Modeled paramotor-UAV impact dynamics under stochastic touchdown; presented first-author at 2026 AIAA Aviation Forum.
- Derived a 16 mode hybrid system of differential equations for the UAV's roll, pitch, and vertical motion during impact.
- Built a C++/OpenMP Monte Carlo over 5,000 stiffness-damping configs, sizing a suspension to a 25% structural force margin.

## Projects

**Autonomous Maze Navigating Robot** | *C++, Arduino* — *Sep 2023 - Jun 2025*

- Implemented cascaded PID (encoder velocity/position; gyro heading) with sensor fusion of odometry and IMU to limit drift.
- Built A* across multi-dimensional state space for $6\times7$ maze with 5-gate sequencing and direction-aware turn costs.
- Engineered trapezoidal motion profiles with scaled heading correction, stall detection, and kick recovery at a 1kHz loop rate.
- **Result:** 3rd place at the 2024 Science Olympiad National Tournament.

**NJ High Injury Network Generator** | *Python, FastAPI, PostgreSQL/PostGIS* — *March 2026*

- Built full-stack web tool that identifies statistically significant high-crash corridors for NJ municipal safety planning.
- Integrated NJDOT crashes with OpenStreetMap road geometries in PostGIS to aggregate crashes by segment.
- Flagged high-crash corridors across 568 NJ municipalities statewide by applying Poisson tests to 5 open data sources.

## Technical Skills

**Languages:** Python, Java, C++, C, HTML/CSS/JavaScript, Go, OCaml, Rust, SQL, Typscript, LaTeX
**Libraries:** NumPy, SciPy, Scikit-learn, pandas, Matplotlib, TensorFlow
**Tools:** Git, Jupyter, Linux, Bash, Optuna, REST APIs, Kubernetes, Docker, Jira, CI/CD, Grafana
**Technical Areas:** Monte Carlo Simulation, ODE Modeling, Surrogate Modeling, PID Control, Sensor Fusion, Path Planning